Anal Roy Chowdhury

Ph.D. Student · School of Mathematical & Computational Sciences · IACS, Kolkata

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Education

Ph.D., Mathematical & Computational Sciences, IACS Kolkata — Aug 2023 – Present
Supervisor: Dr. Debarshi Kumar Sanyal · Focus: Multimodal Learning, Sign Language Processing

M.Sc., Electronic Science, Acharya Prafulla Chandra College (WBSU) — 2016 – 2018
Thesis: Electrical Properties of ID-DG & TG-DG MOSFETs (Ortiz-Conde Model) · Supervisor: Dr. Arpan Deyasi

B.Sc., Physics (Hons.), Seth Anandram Jaipuria College, University of Calcutta — 2013 – 2016

Research Experience

Ph.D. Research Scholar · SMCS, IACS Kolkata — Aug 2023 – Present
- Building motion-aware and vision–language deep learning systems for Indian Sign Language (ISL) recognition and translation.
- Published at **ACL 2026** (Findings) and **WSLP @ AACL-IJCNLP 2025**.

Project Linked Personnel · ISI Kolkata — Jul 2021 – Mar 2023
- Investigated multiple kernel methods for multi-view clustering under Dr. Swagatam Das (ECSU); led a comprehensive survey of 100+ deep multi-view clustering methods.
- Survey published in **Information Fusion** (Elsevier, 2025).

Research Project Assistant · Sangutech Enterprises Pvt. Ltd., DBT-BIRAC funded — Feb – Dec 2019
- Developed deep learning pipelines for biomedical image analysis; co-invented UV-light-based platelet aggregation detection system (Multi-scale FCN).
- Resulted in **Indian Patent IN580844 B1** (granted 17 Feb 2026).

Selected Publications

[1] **A. Roy Chowdhury**, A. Gupta, S. Das. "Deep multi-view clustering: A comprehensive survey." *Information Fusion*, Elsevier, 2025. [DOI]

[2] **A. Roy Chowdhury** and D.K. Sanyal. "Can Small Vision–Language Models Perform Sign Language Translation?" *Findings of ACL 2026*, San Diego.

[3] **A. Roy Chowdhury** and D.K. Sanyal. "Enhancing Indian Sign Language Translation via Motion-Aware Modeling." *WSLP @ AACL-IJCNLP 2025*, IIT Bombay, pp. 39–46.

[4] A. Deyasi, **A.R. Chowdhury**, K. Roy, A. Sarkar. "Effect of High-K Dielectric on Drain Current of ID-DG MOSFET." *IEEE EDKCON 2018*, pp. 176–181. [DOI]

[5] **A.R. Chowdhury**, K. Roy, P. Saha, A. Deyasi. "Tunneling Current in Nano-MOSFET Using BSIM4 Model." *CCIS*, Springer, vol. 836, pp. 11–18, 2018. [DOI]

Patent

[P1] S. Deb, R. Chowdhury, A. Sadhu, **A. Roy Chowdhury**, A.K. Dasgupta, A.K. Dhara, K. Roy, M. Bhattacharyya, A. Chakrabarti. "Agonist-free system and method for detecting platelet function." **Indian Patent IN580844 B1**, granted 17 Feb 2026. App. No. 201931048635.

Technical Skills

Languages	Python, MATLAB, C
Frameworks	PyTorch, Scikit-learn, OpenCV
Research	Multi-view Learning, VLMs, NLP
Tools	Git, Jupyter, LaTeX

Awards & Service

- **UGC-NET (LS)** — Electronic Science (Dec 2020, Jun 2021 merged cycle)
- **Travel Grant** — CODS-COMAD 2024 (IIT Jodhpur) & 2023 (IIIT Bangalore)
- **Admin Chair**, WSDL 2023, ISI Kolkata; **Organising Member**, WSDL 2022